- Historical average timeline from discovery to first production is often cited around **15-17 years**, and even from a firm "investment decision" (final feasibility/construction start) to first production typically takes **5-10+ years** for a major sulphide project, longer if permitting is contested (as in Peru, Panama).
- Industry estimates: >**$210 billion** in new investment needed by 2035 to meet projected demand, nearly 3x the ~~$76B deployed in the prior six years. Capex per tonne of new capacity has risen materially (~~40-65% increases cited for new Latin American supply since 2020) due to lower grades, deeper/harder ore bodies, water/energy constraints, and permitting costs.
- **Implication for a quant model**: because new mine supply cannot respond to price signals within a 1-22 trading day (or even 1-2 year) horizon, short-term copper price moves are almost entirely a **demand-and-inventory-and-shock** story, not a mine-supply-response story, except through the narrow channel of unplanned outages (strikes, weather, accidents) hitting an already-inelastic supply curve. This is why supply-side news (a strike, a mudslide) moves price disproportionately relative to its % of global tonnes — there's no slack.
- **Timing:** structural/leading only at multi-year horizon. **Evidence:** QUANT (capex/production data exist) but this is a slow variable, useful mainly as a *prior* on how tight the supply curve is, not a forecasting feature at daily/monthly horizon. **Data:** PAID for granular project-level tracking (Wood Mackenzie, CRU, S&P Global); FREE approximations via USGS annual data and company financial filings/press releases.

#### 1.4 TC/RCs (treatment and refining charges) — genuinely useful leading indicator of concentrate tightness

- TC/RCs are the fees smelters charge miners to convert concentrate to refined metal. They fall when concentrate is scarce (smelters compete for feedstock, accept lower fees) and rise when concentrate is abundant.
- 2025-2026 status: spot TCs have gone **negative** (as low as ~-$60/tonne in Nov 2025) for the first time at this severity — a signal that Chinese smelting capacity has outgrown mined concentrate supply. This is described by multiple sources as the most acute concentrate-scarcity signal in over a decade.
- **Mechanism relevance for price**: TC/RC tightness is a leading indicator of *smelter margin stress and potential smelter production cuts*, which eventually feeds through to refined-metal supply (with a lag of months, since smelters typically only cut output after sustained negative margins). It is NOT a same-day price driver; it's a structural/medium-term (weeks-to-quarters) leading indicator of refined supply direction.
- **Timing:** LEADING (for refined supply, on a multi-month lag). **Evidence:** QUANT/directionally well-established mechanism but no widely published backtested "TC/RC → price" regression coefficient found in this research; mostly qualitative/analyst commentary. Treat as MIXED. **Data:** PAID for real-time spot TC/RC series (Fastmarkets, SMM, S&P Platts); quarterly benchmark TC/RCs are reported in free news but with a lag and low frequency (quarterly negotiations, e.g. the annual China benchmark deal).

#### 1.5 Smelter capacity/utilization

China dominates global smelting; smelter maintenance schedules, environmental-inspection shutdowns, and negative-TC-driven curtailments are a real but hard-to-quantify-in-real-time supply lever. **Data:** mostly PAID (SMM, Antaike, Fastmarkets China coverage); some free news coverage of major curtailment announcements.

#### 1.6 Scrap copper supply

- Scrap (secondary supply) is roughly 15-20% of total copper supply and is more price-elastic/cyclical than mined supply — high prices pull more scrap into the market with a lag of weeks to a couple months (collection/processing response), while scrap generation also tracks the replacement cycle of end-of-life products (cars, wiring, appliances) with roughly a multi-year lag to the original consumption boom.
- Recent evidence (2025-2026) shows scrap supply has been *slower* to respond to high prices than expected, due to quality/availability constraints and export policy (e.g., China scrap import rules), meaning the traditional "high price pulls in scrap, caps the rally" dampening mechanism has been weaker than historical norms — worth watching as a regime-dependent effect rather than a constant.
- **Timing:** COINCIDENT-to-slightly-LAGGING response to price (scrap flow reacts to price, doesn't lead it), but *scrap flow itself* can be a short-term supply-side variable worth monitoring for its dampening effect on rallies. **Evidence:** MIXED — academic literature exists on refined-scrap price linkages (cointegration studies) but real-time scrap flow data is sparse and lagged. **Data:** mostly PAID (ISRI, Fastmarkets scrap indices, SMM); China customs trade data (scrap import volumes) is FREE but monthly and lagged.

#### 1.7 Labor disputes / strikes — recurring, quantifiable supply shocks

See Section 5 for specific case studies (Escondida 2017, 2024; Grasberg 2025; Cobre Panama 2023; Las Bambas recurring blockades). General pattern: strikes at a top-5 mine reliably produce a **multi-percent single/multi-day price spike**, decaying over subsequent weeks as the market prices in the actual duration/severity of the outage. **Timing:** genuine SHOCK (unpredictable in exact timing, though labor contract renewal dates and strike-vote news are a short-lead-time warning, typically days to a few weeks). **Evidence:** QUANT — price moves around specific strike dates are well documented in news/market data, event-study-able. **Data:** FREE (news, company press releases, contract expiration calendars are public).

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## 2. Demand Side Structure

#### 2.1 China dominance

China consumes roughly **50-60%** of global refined copper. This makes China-specific data far more decision-relevant than global aggregates.

Key China demand sub-drivers:

- **Property/construction sector**: historically one of the largest copper end-uses in China (wiring, plumbing, air conditioning) but has been in structural decline since the 2021-2023 property crisis (Evergrande etc.); property's *share* of Chinese copper demand has been shrinking, offset by grid and EV/renewables growth. A researcher should track Chinese new-construction-starts / property-investment data (available via China National Bureau of Statistics, though data-quality caveats apply to Chinese official statistics) as a *declining-weight* but still relevant input.
- **State Grid investment**: State Grid Corp of China has announced dramatic increases in spending — ~~4 trillion yuan (~~$574B) over 5 years per recent plans, with 2025 annual spending over 650bn yuan (up from 600bn in 2024), and reports of a "40% surge in investment through 2030." Grid buildout (UHV lines, smart substations, transformers) is copper-intensive and increasingly policy-driven rather than cyclical — meaning it can be a demand floor that's *less correlated with the general business cycle* than

construction or manufacturing. **Timing:** grid capex plans are typically announced/budgeted well ahead (LEADING, months-to-years visibility) since they're government five-year-plan items. **Data:** State Grid/NDRC announcements are public and covered in trade press (FREE, though official granular spending data can be PAID via specialized China research services).

- **EV production**: an EV uses roughly **3-5x the copper of an ICE vehicle** (commonly cited figures: ICE ~23-24kg, BEV ~80-91kg, hybrids ~40-60kg), concentrated in battery current collectors, busbars/high-voltage cabling, and motor windings. China EV production/sales data (CPCA monthly sales figures) is timely, free, and a genuine leading proxy for a growing demand bucket. **Timing:** LEADING relative to physical copper offtake by weeks (production precedes wiring-harness/component purchasing by a short lag). **Evidence:** QUANT on copper-intensity-per-vehicle (well-established engineering figures); less rigorously connected in public literature to precise price elasticities. **Data:** FREE (CPCA, CAAM monthly China EV data; EV-Volumes, IEA Global EV Outlook globally).
- **Renewables (solar/wind) buildout**: solar PV inverters ~370 kg copper/MVA; wind PMSG converters ~519 kg copper/MVA; onshore wind turbines use roughly 10 tonnes of copper each (offshore more than double). Renewable generation broadly requires an estimated **2.5-7x more copper per unit of output** than fossil generation (cabling, transformers, inverters). China solar installation data is exceptionally fast-growing and volatile (China added ~198GW of solar in just the first 5 months of 2025, +150% YoY) — this is a genuinely large, trackable, monthly-frequency demand driver. **Timing:** roughly COINCIDENT-to-slightly-LEADING (installation data precedes full grid-tie-in copper demand by weeks). **Data:** FREE (China NEA monthly installation data, BNEF/IEA aggregates — though BNEF detailed data is PAID).

### 2.2 Global manufacturing PMI as demand proxy

- Global and China (Caixin/official) manufacturing PMI show a **decent, time-varying correlation with copper price** — correlation reportedly dropped near zero in 2022 (distorted by the Russia/Ukraine shock and China COVID lockdowns) before re-strengthening. Copper-specific "copper users PMI" series (tracked e.g. by MacroMicro) are more tightly correlated than headline PMI.
- Industrial-production-index correlation with copper is cited around **0.65-0.75** in some analyses.
- **Timing:** PMI is typically treated as COINCIDENT-to-slightly-LEADING for near-term industrial activity (new orders sub-index has more lead than headline), and it's published monthly with only a few days' lag — genuinely fast and usable. **Evidence:** MIXED-to-QUANT — correlation is real but non-stationary/regime-dependent (breaks down during exogenous shocks), so treat as a moderately reliable feature, not a standalone signal. **Data:** FREE (ISM US PMI, S&P Global/Caixin China PMI headlines are free; sub-index and country-level detail sometimes requires subscription).

### 2.3 Seasonality

Copper demand has mild, well-known seasonal patterns: Chinese New Year (Jan/Feb) causes a demand/consumption lull and inventory buildup at SHFE as factories close, followed by a post-holiday restocking demand pulse (March-April, the classic "spring restocking" narrative in Chinese commodity markets); Northern Hemisphere construction activity picks up in spring/summer. These are real, recurring, low-magnitude patterns rather than major price drivers — more useful for interpreting inventory data (e.g., a SHFE stock build in February is seasonally "normal," not necessarily bearish) than as a standalone forecasting feature. **Timing:** LEADING/calendar-known in advance. **Evidence:** LORE-to-MIXED (widely discussed by market participants, less rigorously backtested in public academic literature). **Data:** FREE (SHFE stock data has strong day-of-year seasonality visible directly in the historical series).

(SHFE stock data has strong day-of-year seasonality visible directly in the historical series).

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## 3. Inventory Dynamics

### 3.1 The three visible exchange stocks

- **LME** (London Metal Exchange) warehouse stocks — global network, includes historically significant Asian (esp. Korea/Taiwan/Malaysia) warehouse locations; daily reported, free.
- **COMEX** warehouse stocks — US-centric, daily reported, free (CME Group website).
- **SHFE** (Shanghai Futures Exchange) — weekly reported (Fridays), domestic China-only stocks, free.

### 3.2 The "invisible"/bonded inventory problem — a real, documented data-quality issue

This is a genuine and important caveat for a quant researcher: **exchange-visible stocks are known to be an incomplete and sometimes misleading measure of true global copper inventory.**

- China's **bonded warehouse** stocks (held in Shanghai/other bonded zones, outside SHFE's official reporting, often for financing/arbitrage purposes — copper used as loan collateral in China's shadow-financing system) are estimated at roughly **500,000 to 1,000,000+ tonnes** at various points, but are privately held and not officially published. The International Copper Study Group (ICSG) and private consultancies (e.g., SMM) produce *estimates*, not hard data, often reconciling three independent consultant estimates.
- These invisible stocks move in and out of official exchange registration depending on financing costs, price arbitrage between SHFE and LME, and China's capital controls/financing conditions — meaning a big draw on SHFE-visible stock can sometimes just reflect a *shift from bonded to registered* rather than genuine consumption, and vice versa.
- **Bottom line for the researcher**: visible exchange inventory (LME+COMEX+SHFE combined, sometimes called "on-exchange" stock) is a real, timely, free signal, but it is a **noisy and partial proxy** for true global copper stock levels — known and openly discussed problem in the professional copper analyst community, not folklore. Treat inventory-level (the stock, in absolute tonnes) with more caution than inventory *changes/draws* over short windows, and be aware that China arbitrage flows (the SHFE-LME price spread driving physical relocation) can distort the raw draw/build number independent of underlying consumption.

### 3.3 Do inventory draws/builds have short-term predictive power?

- The qualitative/practitioner view (found across multiple sources) is consistent: sharp draws reinforce bullish price narratives, persistent builds reinforce bearish ones, and *markets react more to the rate of change and the change relative to the (seasonally adjusted) expected level* than to the absolute stock number ("relative inventory changes... weighted... against absolute inventory levels").
- Academic literature (LASSO/EMD decomposition studies, SVAR/structural studies of the copper market, convenience-yield/inventory-based commodity pricing models in the finance literature more broadly — e.g., the classic theory of storage) supports inventories as one of several statistically significant explanatory factors in copper price models, generally bundled with macro and financial variables rather than isolated as a standalone strong predictor.
- No single, widely-cited, easily reproducible published coefficient (e.g., "a 10% weekly SHFE draw predicts an X% price move over Y days") was found in this search — the evidence supports inventories

predicts an X% price move over Y days") was found in this search — the evidence supports inventories mattering, but the *precise, backtested, out-of-sample predictive power at a 1-22 day horizon* is thinner than the strength of the qualitative narrative. **Evidence: MIXED** — real mechanism (theory of storage / convenience yield is well-established finance theory), genuine practitioner usage, but weak/sparse rigorous public quantification specific to copper at short horizons.

- **Timing:** COINCIDENT-to-slightly-LEADING (a draw reflects consumption that already happened, but a *persistent trend* of draws can lead further price appreciation if it signals an emerging structural deficit).
- **Data:** FREE and high-frequency (LME/COMEX daily, SHFE weekly) for on-exchange stock; bonded/invisible stock estimates are PAID or semi-public via trade press summarizing ICSG/SMM estimates (monthly, lagged).

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## 4. Price Relationships / Cross-Asset Signals

### 4.1 "Dr. Copper" — copper as a leading macro indicator

- There IS real academic support, not pure folklore: a study using monthly data Jan 1990-June 2023 found statistically significant **unidirectional Granger causality from copper prices to the U.S. Industrial Production Index** (at a one-month lag). Other academic work finds copper consumption and economic growth are cointegrated with long-run causality running (in panel studies of rich countries) from growth to consumption — i.e., the causality direction is genuinely debated in the literature and likely bidirectional/regime-dependent.
- Practically: copper's leading property is strongest for *global industrial activity/manufacturing cycle turning points*, weaker as a predictor of broader GDP/recession calls (where its track record has some well-known high-profile misses, and financial press pieces have explicitly begun to question the "Dr. Copper" moniker in recent cycles as demand structure shifts toward China-policy-driven grid/EV spending, decoupling copper somewhat from the general global business cycle).
- **Timing:** LEADING (weak-to-moderate) for industrial production at ~1-month lag. **Evidence:** MIXED — genuine Granger-causality evidence exists (QUANT), but the popular "Dr. Copper predicts recessions" framing is broader than what the specific studies support (partially LORE). **Data:** FREE (industrial production indices are FRED data; copper price is free).

### 4.2 Copper-Gold ratio

- Widely used as a risk-on/risk-off and growth-expectations gauge; historically correlated with the 10-year Treasury yield at roughly **0.85 over 2000-2020** per cited research, and there's a peer-reviewed paper (ScienceDirect) specifically framing the copper-gold ratio as a **leading indicator for the 10-year Treasury yield.**
- Important caveat found in multiple 2024-2026 sources: this relationship has been **breaking down/decoupling post-pandemic** — China's slowdown pulled the ratio down while yields rose on inflation/Fed tightening, decoupling the two series. This is a good example of a historically strong relationship whose stability a researcher should NOT assume is constant — worth using as a *regime-dependent* feature (e.g., interact with a China-macro-surprise variable) rather than a static linear predictor.
- **Timing:** the ratio itself is coincident with copper price (definitionally, since copper is one leg); its value is as a cross-check/regime indicator alongside other macro series, and per the cited paper, LEADING

specifically vs. the 10Y yield. **Evidence:** QUANT (peer-reviewed correlation study exists) but non-stationary relationship — MIXED overall. **Data:** FREE (copper and gold prices are both trivially available; ratio is a simple derived series).

**4.3 Copper vs. US Dollar Index (DXY)**

- Strong, consistent negative correlation: multiple sources cite rolling 20-year correlation coefficients in the **-0.65 to -0.82** range, strengthening to **-0.90 or beyond** during high-volatility periods; one source cites a rule-of-thumb "1% DXY rise ≈ 0.3-0.7% base-metals decline."
- Mechanism is well understood (commodities are dollar-denominated globally; dollar strength mechanically raises the cost for non-dollar buyers and also generally coincides with tighter global financial conditions that suppress commodity demand/speculative positioning). This is one of the more robust, mechanically-grounded, and consistently-documented relationships in the whole list.
- **Timing:** best treated as COINCIDENT/contemporaneous (DXY and copper often move within the same session on macro news), though DXY momentum/trend can have some short lead value via its correlation with real-rate and Fed-policy expectations which are set slightly ahead of commodity repricing. **Evidence:** QUANT — this is the most rigorously and consistently documented cross-asset relationship in this research. **Data:** FREE (DXY is on FRED/Yahoo Finance, updated continuously).

**4.4 Copper vs. other industrial metals (Al, Zn, Ni)**

- High, but not perfect, co-movement: copper-aluminum correlation cited at ~0.87 (2011-2021); zinc-copper/zinc-aluminum correlation ~0.6-0.8 on a rolling one-year basis since 2003 (weaker, ~0.2-0.6, pre-2003). Driven mainly by shared demand exposure (China/EM industrial cycle) rather than direct substitution in most end-uses (copper and aluminum ARE partial substitutes in some applications — power cables, some HVAC/motor windings — which adds a substitution-driven co-movement channel on top of the shared-demand-cycle channel).
- Practical use: correlated base-metal price/return series (or a basket average) can serve as a noise-reduction/confirmation signal or as a control variable to separate "copper-specific" shocks (mine strikes, TC/RC news) from "industrial-metals-complex-wide" macro shocks (China PMI surprise, global growth scare). A copper move that is NOT confirmed by aluminum/zinc/nickel moving the same direction is more likely idiosyncratic (mine supply news) than macro-driven, which is itself a useful classifier feature.
- **Timing:** COINCIDENT. **Evidence:** QUANT (correlation studies, spillover/connectedness academic literature e.g. Springer Mineral Economics 2025 paper on time-varying spillovers). **Data:** FREE (LME metal prices widely available; nickel has its own idiosyncratic risk post the 2022 LME nickel short squeeze, worth excluding/flagging as an outlier period if used in a model).

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## 5. Historical Price-Shock Case Studies

| # | Event | Date(s) | Magnitude | Timeframe | Predictable in advance? |
|---|---|---|---|---|---|
| 1 | Global Financial Crisis collapse | Apr–Dec 2008 | ~69% crash ($3.91/lb → $1.29/lb) | ~8 months | Macro/credit stress was visible for months prior (genuine leading indicators; credit spreads |

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| | collapse | | | | indicators: credit spreads, LIBOR-OIS), but the magnitude/speed of the commodity-specific collapse was a shock amplified by deleveraging/margin calls, not purely fundamentals-driven |
| 2 | Post-GFC China-stimulus rally | 2009–Feb 2011 | Rally to all-time high ~$4.65/lb | ~2 years | Partially predictable once China's ~4 trillion yuan stimulus package was announced (Nov 2008) — a genuine leading policy signal, though the size/duration of the rally still surprised many |
| 3 | Escondida strike | Feb–Mar 2017 | 44-day strike; ~5% of global mined supply offline; price spiked toward/above $6,000/t (multi-% single-week moves) | ~6 weeks | Partially predictable — strike-vote news and contract-expiration dates were public days/weeks ahead, giving genuine short lead time (a "labor calendar" feature) |
| 4 | Cobre Panama shutdown | Nov 2023 (Panama Supreme Court ruling; ordered closure) | Removed ~1.5% of global supply / 40% of First Quantum's revenue; contributed to a multi-month tightening in the physical market and was a factor behind 2024's copper price strength | Ongoing (still shut as of this research; ~5% of Panama's GDP impact) | Genuine shock — the speed of the Supreme Court's unconstitutionality ruling and government closure order was not something a price model could have anticipated from market data; political/legal risk, not a market-signal-detectable event |
| 5 | COVID-19 crash and China-stimulus/EV rally | Mar 2020 (crash) → 2020-2022 (rally to new highs ~$4.90+/lb by 2022) | Crash of ~25-30% in weeks (Mar 2020), then >100% rally over ~2 years | Weeks (crash) / ~2 years (rally) | Crash was a genuine, fast exogenous shock (undetectable from copper-specific data alone — a cross-asset liquidity/pandemic shock); the subsequent rally was more |

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| | | | | | predictable in direction (given announced Chinese stimulus and global fiscal responses) though magnitude was still surprising |
| 6 | Grasberg mudslide / force majeure | Sept 8, 2025 (event); force majeure declared ~Sept 24, 2025 | Price surged to 15-month highs, single-day move ~+2.7% on the announcement day, with ~2-3% of global mined supply affected and disruption extending into mid-2026 per company guidance | Days (initial spike) to over a year (supply recovery) | Genuine shock — geological/accident risk not foreseeable from market data; however, once the initial incident occurred, the force-majeure declaration and subsequent guidance provided a short, tradeable lead window between "incident news" and "market fully re-pricing" |
| 7 | Las Bambas / Peru community blockades | Recurring, various dates 2022-2025 | Individually smaller than the above (typically low-single-digit % production impact each time) but recurring frequently enough to be a structural "Peru risk premium" | Days to weeks per episode | Partially predictable — recurring pattern tied to specific communities/roads, some advance warning via local news, but exact timing of each blockade is a shock |
| 8 | China property crackdown ("three red lines" policy) and subsequent property crisis | Aug 2020 policy announcement; Evergrande crisis unfolding through 2021-2023 | Contributed to a multi-year drag/decoupling of the construction-demand component of Chinese copper demand, though offset by grid/EV growth — more of a structural demand-composition shift than a single sharp price shock | Multi-year | Policy announcement itself (Aug 2020) was a genuine, publicly visible LEADING signal — a clear example of Chinese policy news functioning as a leading indicator ahead of the demand impact showing up in physical data |

price shock

**General pattern for the researcher**: mine-specific supply shocks (strikes, accidents, political shutdowns) are the most genuinely *unpredictable* (Type: shock) category but often have a short public-information lead window (strike votes, court proceedings, force-majeure announcements) that a news/text-based feature could partially capture even if the underlying event itself can't be forecast. Macro/policy-driven moves (China stimulus, property policy, Fed tightening cycles) are comparatively more foreseeable from publicly announced policy calendars and macro data, though magnitude is still routinely underestimated by the market.

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## 6. Futures Market Structure

### 6.1 Contango / backwardation

- Backwardation (near-term/spot prices above deferred futures) signals tight physical supply / strong immediate demand; contango (deferred above spot) signals ample supply / high carrying-cost environment. For copper specifically, the curve has swung between the two states depending on inventory and China demand conditions.
- There is real research (cited: LME base-metals studies finding financial variables, activity proxies, and the futures basis itself help predict futures/spot returns) supporting genuine short-term predictive content in the basis, consistent with the broader academic "theory of storage"/convenience-yield literature in commodity finance (a well-established, decades-old body of academic work, not new to copper specifically).
- **Timing:** the curve shape is essentially COINCIDENT with the current physical tightness, but changes in the curve shape (steepening backwardation) can lead spot price appreciation by days-to-weeks as it reflects real-time physical scarcity before it's fully reflected in headline spot. **Evidence:** QUANT — commodity futures basis-predicts-returns is one of the more rigorously studied relationships in commodity finance broadly (theory of storage, Fama-French commodity basis papers, etc.), reasonably transferable to copper. **Data:** FREE (LME/COMEX/SHFE forward curves are published; front-month vs. deferred spreads computable from daily settlement data).

### 6.2 COT / speculative positioning

- CFTC weekly Commitments of Traders report breaks out "Managed Money" (hedge funds/CTAs) net positioning in COMEX copper futures — the category most relevant as a sentiment/positioning gauge.
- Known data limitation: COT data is a **snapshot as of the prior Tuesday, released the following Friday** — a real reporting lag (3 days minimum, effectively up to a week old by the time it's stale-dated against subsequent price action) that meaningfully limits its use as a short-horizon (1-22 day) predictive feature versus its use as a *contrarian/extremes* indicator (e.g., "managed money net-long positioning at a multi-year extreme" is more of a mean-reversion / crowding signal than a direction predictor).
- The general finding across commodity-positioning literature (broader than copper specifically) is that extreme speculative positioning tends to be **COINCIDENT-to-LAGGING** relative to price (speculators chase trends rather than lead them), useful mainly as a crowding/reversal-risk indicator rather than a standalone directional signal. No copper-specific rigorous predictive-power study was found in this search; treat as MIXED-to-LORE for direct directional forecasting, though legitimately useful as a

search, treat as MIXED-to-LOW for direct directional forecasting, though legitimately useful as a positioning/crowding *risk* feature (e.g., flag elevated reversal risk when extreme).

- **Data:** FREE (CFTC.gov publishes COT reports weekly at no cost — one of the better free/quant-friendly data sources in this whole list, despite the lag).

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### Ranked List: Factors With the Most Genuine Short-to-Medium-Term (1-22 Trading Day) Predictive Signal

*This ranking is my own judgment, synthesizing the evidence/timing/data-availability assessments above — not a market consensus. Ranked roughly by (real evidence for predictive value) × (data is genuinely obtainable and sufficiently timely for a 1-22 day horizon).*

1. **On-exchange inventory changes (LME + COMEX + SHFE combined, rate-of-change not level)** — free, daily/weekly, genuine mechanism (theory of storage), well-known limitation (bonded stock noise) is itself modelable by tracking the SHFE-LME arbitrage spread as a confound variable.
2. **US Dollar Index (DXY) level and momentum** — the single most robust, mechanically-grounded, consistently-documented relationship found in this research (-0.65 to -0.90 correlation across studies); free and real-time.
3. **Futures curve shape / near-term calendar spread (backwardation vs. contango, and its rate of change)** — grounded in well-established commodity-finance theory (basis predicts returns), free data, directly computable, appropriately short-horizon.
4. **China copper-relevant PMI (Caixin manufacturing PMI, copper-users PMI where obtainable)** — monthly frequency limits it to a lower-frequency feature/regime variable rather than a daily signal, but genuinely leading for the demand cycle and free.
5. **TC/RC levels and trend (spot, if obtainable; quarterly benchmark otherwise)** — strong underlying mechanism as a concentrate-tightness leading indicator for refined supply; main weakness is real-time spot data is paid and lower-frequency, but even the directional trend (deepening negative TC/RCs through 2025) is a genuine multi-month leading signal.
6. **Mine-supply-disruption news flow (strikes/force-majeure/accidents at the ~15-20 largest mines), as an NLP/event-detection feature** — the single highest-magnitude, fastest (same-day) price-moving factor category demonstrated in the case studies (Escondida, Grasberg, Cobre Panama), free via news, and well-suited to a text/event-extraction pipeline given a maintained watchlist of major mines and their labor-contract calendars.
7. **Copper-gold ratio and its divergence from trailing norm** — real academic backing as a cross-asset regime signal, free data, but treat as regime-dependent (explicitly flagged as decoupling in recent years) rather than a stable linear predictor — use as a conditioning/regime variable, not a standalone forecast input.
8. **Correlated industrial-metals basket (aluminum, zinc; nickel with caution post-2022 squeeze) return, as a "macro-vs-idiosyncratic" decomposition tool** — free, high-frequency, doesn't predict copper directly but is genuinely useful for classifying whether a copper move is macro-driven (confirmed by the basket) or copper-specific (a mine/TC event), which materially changes the appropriate model response.
9. **China EV production/sales (CPCA monthly) and China NEA solar/wind installation data** —

structurally growing, policy-supported demand components that are increasingly decoupling from the traditional PMI/construction cycle; monthly frequency limits short-horizon use but valuable as a slow-moving demand-floor/trend feature and for interpreting whether "soft PMI" still means "soft copper demand" in a given period.

10. **CFTC COT managed-money net positioning, used as a crowding/extremes flag rather than a direction predictor** — free, weekly, real lag limitation, best used as a risk/reversal-probability feature (e.g., a volatility or mean-reversion regime flag) rather than a standalone directional signal.

**Explicitly de-prioritized / lower-confidence for this specific horizon, despite popularity:**

- "Dr. Copper predicts recessions/GDP" — the specific, rigorous evidence (Granger causality to industrial production) is real but narrower and weaker than the popular framing, and monthly/lower-frequency besides.
- Scrap supply flow — real mechanism, but data is largely paid, lagged, and recent evidence (2025) suggests the relationship has been weaker/less reliable than historically assumed.
- Ore grade decline / mine capex cycle data — real and important for understanding *why* supply is structurally inelastic (useful as background/prior for how much weight to give supply-shock news), but far too low-frequency to function as a forecasting feature at a 1-22 day horizon.

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## Data Source Quick Reference

| Category | Free sources | Paid sources |
|---|---|---|
| Prices (spot, futures, curves) | LME, CME/COMEX, SHFE websites; Yahoo Finance; Investing.com | Bloomberg, Refinitiv Eikon |
| Inventory (on-exchange) | LME, CME, SHFE daily/weekly stock reports | — |
| Inventory (bonded/invisible) | Trade-press summaries of ICSG estimates (lagged, low-frequency) | SMM, Fastmarkets, ICSG full dataset |
| TC/RC | Quarterly benchmark deals reported in trade press | Fastmarkets, S&P Platts, SMM (spot/real-time) |
| Macro (PMI, industrial production, DXY, yields) | FRED, ISM, S&P Global/Caixin PMI headlines, Yahoo Finance | Detailed sub-indices via S&P Global subscription |
| Positioning (COT) | CFTC.gov (weekly, free) | — |
| China demand detail (EV, solar/wind, grid, property) | CPCA, CAAM, China NEA, NBS (official stats, free but data-quality caveats) | BloombergNEF, Wood Mackenzie, CRU for granular/verified detail |
| Mine-level supply/production/capex | Company quarterly reports, USGS annual summaries | Wood Mackenzie, CRU, S&P Global Market Intelligence |